COUNTEREXAMPLES TO THE CHAN–PAK–PANOVA MATCHING CONJECTURE

Abstract. Chan, Pak, and Panova proposed a promotion–demotion bipartite graph whose conjectured matching would give a direct injection for Stanley's log-concavity inequality. Using the inverse-order word sweeps that realize the stated promotion–demotion effect, we disprove their Conjecture 9.4. For every $m \geq 2$, let P_m have elements $x, d, b_1, \dots, b_m$ and only nontrivial comparability $x \prec d$, and take $a = 2$. The resulting graph G_m satisfies

$$|V| = m^2(m!)^2, \qquad |W| = (m^2-1)(m!)^2, \qquad \nu(G_m) = |\mathcal{N}(W)| = (m-1)(m!)^2.$$

Thus every maximum matching leaves $m(m-1)(m!)^2$ vertices of W unmatched, even though $|V| - |W| = (m!)^2 > 0$. The smallest counterexample has three elements and is the antichain; the four-element poset P_2 is smallest among counterexamples for which Stanley's inequality is strict.

1. The matching conjecture

Let $P = (X, \prec)$ be a finite poset, let $x \in X$, and let $\mathcal{E}(P, x, r)$ be the set of linear extensions placing x in position r. Stanley's inequality [2] states that

$$|\mathcal{E}(P,x,a)|^2 \geq |\mathcal{E}(P,x,a-1)|\,|\mathcal{E}(P,x,a+1)|. \tag{1}$$

Chan, Pak, and Panova [1, Conjecture 9.4] proposed a bipartite graph with

$$V = \mathcal{E}(P,x,a) \times \mathcal{E}(P,x,a), \qquad W = \mathcal{E}(P,x,a-1) \times \mathcal{E}(P,x,a+1),$$

and conjectured that it has a matching saturating W; in the wording of [1], that there exists a maximal matching covering all vertices in W.

We make the promotion–demotion convention explicit. Write a linear extension as a word $w = w_1 \cdots w_n$. For $1 \leq r < n$, let τ_r interchange w_r and w_{r+1} when they are incomparable, and fix w otherwise. Section 7 of [1] uses a right action, whereas Section 9.12 writes the operators on the left. We use the inverse-order word sweeps that realize the stated effect in Section 9.12:

$$D_i(w) = w\tau_{i-1} \cdots \tau_1, \qquad U_j(w) = w\tau_j \cdots \tau_{n-1}. \tag{2}$$

Equivalently, these are induced by the left action $g \cdot w := wg^{-1}$. For $(\pi, \sigma) \in W$ and an element y occurring after x in π and before x in σ, let i and j be the positions of y in π and σ, respectively, and join (π, σ) to

$$(D_i(\pi), U_j(\sigma)) \in V.$$

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For $S \subseteq W$, write $\mathcal{N}(S) \subseteq V$ for its neighborhood, and let $\nu(G)$ denote the maximum matching size.

2. An infinite family

Theorem 1. *For every $m \geq 2$, let*

$$X_m = \{x, d, b_1, \dots, b_m\}$$

and let P_m have only nontrivial comparability $x \prec d$. For the graph G_m above with $a = 2$,

$$|V| = m^2(m!)^2,$$
$$|W| = (m^2 - 1)(m!)^2,$$
$$\nu(G_m) = |\mathcal{N}(W)| = (m-1)(m!)^2.$$

Consequently, under the convention in (2)*, Conjecture 9.4 of* [1] *is false. Every maximum matching leaves exactly*

$$|W| - \nu(G_m) = m(m-1)(m!)^2$$

vertices of W unmatched, while

$$|V| - |W| = (m!)^2 > 0, \qquad \frac{|W|}{|\mathcal{N}(W)|} = m + 1.$$

Proof. Set $B = \{b_1, \dots, b_m\}$ and abbreviate $\mathcal{E}_r = \mathcal{E}(P_m, x, r)$. Since $x \prec d$ is the only nontrivial comparability,

$$|\mathcal{E}_1| = (m+1)!, \qquad |\mathcal{E}_2| = m\, m!, \qquad |\mathcal{E}_3| = m(m-1)(m-1)!.$$

Indeed, if x is first, the remaining elements are arbitrary; if x is second, the first element is any member of B; and if x is third, the first two elements are an ordered pair of distinct members of B. Hence

$$|V| = |\mathcal{E}_2|^2 = m^2(m!)^2$$

and

$$|W| = |\mathcal{E}_1|\,|\mathcal{E}_3| = (m^2 - 1)(m!)^2.$$

For $y \in B$, let $\mathcal{N}_y$ be the set of endpoints in V of edges labeled by y. The two entries before x in every $\sigma \in \mathcal{E}_3$ are distinct members of B, so every edge label belongs to B. Since y is incomparable with every other element, each sweep in (2) moves y through all intervening positions. It follows that $\mathcal{N}_y$ consists exactly of the pairs

$$(yx\alpha,\ zx\beta y), \tag{3}$$

where $z \in B \setminus \{y\}$, α is an arbitrary ordering of $X_m \setminus \{x, y\}$, and β is an arbitrary ordering of $X_m \setminus \{x, y, z\}$. Conversely, every pair in (3) is obtained from

$$(xy\alpha,\ zyx\beta) \in W$$

by the edge labeled y. Therefore

$$|\mathcal{N}_y| = m!\,(m-1)(m-1)!.$$

The sets $\mathcal{N}_y$ are pairwise disjoint, since y is the first letter of the first coordinate and the last letter of the second. Thus

$$|\mathcal{N}(W)| = \sum_{y \in B} |\mathcal{N}_y| = m\, m!\,(m-1)(m-1)! = (m-1)(m!)^2.$$

Finally, the edges

$$(xy\alpha, zyx\beta) \;-\!- \;(yx\alpha, zx\beta y)$$

as y, z, α, β vary are pairwise disjoint on both sides and saturate $\mathcal{N}(W)$. Hence $\nu(G_m) = |\mathcal{N}(W)|$. Since $|\mathcal{N}(W)| < |W|$, Hall's condition fails. The remaining formulas follow immediately. $\square$

The strict inequality $|V| > |W|$ shows that the obstruction persists even when Stanley's inequality (1) is strict. In particular, the result refutes only the proposed matching construction, not Stanley's inequality itself.

3. Smallest counterexamples

Proposition 2. *Up to isomorphism preserving the distinguished element x, the antichain on three elements is the unique smallest counterexample. Every three-element instance with $W \neq \varnothing$ satisfies $|V| = |W|$. Consequently, P_2 is smallest among counterexamples for which Stanley's inequality is strict.*

Proof. For the antichain on $\{x, b, c\}$ with $a = 2$,

$$\mathcal{E}_1 = \{xbc, xcb\}, \qquad \mathcal{E}_2 = \{bxc, cxb\}, \qquad \mathcal{E}_3 = \{bcx, cbx\}.$$

Thus $|V| = |W| = 4$, whereas

$$\mathcal{N}(W) = \{(bxc, cxb), (cxb, bxc)\}.$$

Hence no matching saturates W. No smaller instance is possible because $1 < a < n$ requires $n \geq 3$.

Now let $|X| = 3$ and suppose $W \neq \varnothing$. A linear extension placing x first and one placing x last imply that x is incomparable with the other two elements. Deleting or inserting x gives bijections between the linear extensions of $P \setminus \{x\}$ and each of $\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_3$. Therefore $|\mathcal{E}_1| = |\mathcal{E}_2| = |\mathcal{E}_3|$, and hence $|V| = |W|$.

If the other two elements are comparable, write them as $b \prec c$. Then each $\mathcal{E}_r$ is a singleton, and the edge labeled b joins (xbc, bcx) to (bxc, bxc). Thus the antichain is the only three-element counterexample. The poset P_2 has four elements and, by Theorem 1, satisfies

$$|V| = 16, \qquad |W| = 12, \qquad \nu(G_2) = 4,$$

so it is smallest in the strict regime. $\square$

References

[1] S. H. Chan, I. Pak, and G. Panova, *Effective poset inequalities*, SIAM J. Discrete Math. **37** (2023), no. 3, 1842–1880; preprint version arXiv:2205.02798v3 [math.CO]. All section and conjecture numbers cited here refer to arXiv:2205.02798v3, in which Conjecture 9.4 is stated in Section 9.12 ("Injections and matchings for the Stanley inequality") and the right action on words is set up in Section 7; the journal version may number these differently.

[2] R. P. Stanley, *Two combinatorial applications of the Aleksandrov–Fenchel inequalities*, J. Combin. Theory Ser. A **31** (1981), no. 1, 56–65.